

KHARTOUM INTL, Sudan

WMO: 627210

Lat: 15.589N

Lon: 32.553E

Elev: 1265

StdP: 14.04

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	58.7	61.0	8.5	8.9	90.7	11.4	10.3	85.8	22.4	76.9	20.9	74.9	12.0	0	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	20.7	109.8	67.2	108.9	67.1	107.3	66.9	78.1	90.7	76.9	90.5	76.1	90.1	9.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	138.4	84.5	73.5	130.6	83.3	71.9	123.6	83.8	42.7	91.2	41.3	89.4	40.4	89.4	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.1	19.9	18.1	DB	54.2	113.0	2.9	1.1	52.1	113.8	50.4	114.5	48.8	115.1	46.8	116.0
			WB	40.7	79.4	1.6	2.3	39.6	81.0	38.6	82.3	37.7	83.6	36.5	85.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	87.4	75.5	80.4	85.4	91.4	95.4	95.4	91.4	88.2	91.1	91.4	84.8	77.8	(6)
(7)		DBStd	7.78	5.97	5.66	5.18	4.34	3.37	3.14	4.20	3.95	3.84	3.34	4.59	5.14	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	2	2	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	13642	791	852	1098	1242	1408	1362	1285	1185	1233	1282	1043	861	(10)
(11)		CDD65	8169	328	432	633	792	943	912	820	720	783	817	593	396	(11)
(12)		CDH74	123446	3419	5277	9010	12928	16151	15788	13049	10558	12231	12942	7917	4176	(12)
(13)		CDH80	80003	1554	2857	5539	8870	11708	11477	8694	6325	8012	8569	4492	1906	(13)
(14)	Wind	WSAvg	10.1	11.2	11.4	10.9	10.0	9.2	9.4	11.1	10.8	8.8	7.9	9.8	10.6	(14)
(15)	Precipitation	PrecAvg	4.7	0.0	0.0	0.0	0.0	0.1	0.1	1.5	2.2	0.7	0.1	0.0	0.0	(15)
(16)		PrecMax	17.6	0.0	0.0	0.0	0.0	1.0	0.6	7.3	9.3	3.5	0.6	0.0	0.0	(16)
(17)		PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.6	0.0	0.0	0.0	0.0	0.2	0.2	1.7	2.6	0.7	0.2	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	100.0	104.0	109.1	111.3	112.7	111.6	108.0	104.9	107.4	106.2	102.4	98.2	(19)
(20)			MCWB	63.3	64.2	66.1	67.7	67.9	66.5	68.2	72.7	69.7	68.6	67.2	65.8	(20)
(21)		2%	DB	95.2	100.1	105.6	109.1	110.7	109.6	106.0	102.2	104.9	104.4	100.2	94.9	(21)
(22)			MCWB	63.9	64.0	64.0	67.4	66.8	67.3	67.9	73.0	70.1	68.4	66.3	64.8	(22)
(23)		5%	DB	91.7	96.9	102.2	106.2	109.0	107.8	104.1	100.2	102.6	103.6	98.5	92.6	(23)
(24)			MCWB	63.3	62.9	63.1	65.5	66.6	67.4	69.0	73.1	70.8	68.4	66.0	64.4	(24)
(25)		10%	DB	88.2	93.6	98.8	104.2	107.2	105.9	102.2	98.0	100.7	101.9	96.5	89.7	(25)
(26)			MCWB	61.7	61.8	61.5	64.5	66.2	67.3	70.1	73.3	71.1	68.5	65.0	63.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.6	68.2	68.4	74.4	75.1	76.5	78.1	79.7	80.1	75.8	71.4	69.6	(27)
(28)			MCD B	90.0	96.4	103.7	93.4	97.7	90.7	90.8	91.0	91.7	91.8	96.6	89.9	(28)
(29)		2%	WB	65.9	65.9	66.3	71.0	73.2	74.4	76.5	78.0	78.5	74.3	69.2	67.3	(29)
(30)			MCD B	90.4	95.2	100.7	99.2	96.9	94.6	90.6	89.9	91.4	91.7	95.8	90.5	(30)
(31)		5%	WB	64.3	64.3	64.6	69.4	71.8	72.8	75.3	76.8	77.2	73.0	67.6	65.8	(31)
(32)			MCD B	89.8	94.1	98.5	101.5	96.8	95.0	90.1	90.0	90.7	91.4	95.0	89.4	(32)
(33)		10%	WB	62.5	62.8	62.9	67.1	70.7	71.8	74.4	76.1	76.0	71.9	66.1	64.2	(33)
(34)			MCD B	87.6	91.4	96.3	101.5	96.6	95.0	90.0	89.5	89.9	91.3	93.8	87.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.6	24.6	25.3	24.8	22.0	20.7	19.0	17.2	18.9	20.1	21.7	22.3	(35)
(36)			MCDBR	25.8	26.6	27.4	26.1	24.2	22.3	20.8	19.7	20.6	21.2	22.2	23.5	(36)
(37)			MCWBR	11.3	11.5	11.3	10.6	9.7	8.4	7.3	5.8	7.3	8.5	9.7	10.3	(37)
(38)		5% WB	MCDBR	24.8	25.4	25.6	23.9	19.6	20.3	18.5	17.5	18.3	19.6	22.1	22.8	(38)
(39)			MCWBR	11.8	12.3	11.6	10.2	7.2	7.8	6.2	5.9	7.0	7.3	10.9	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.423	0.465	0.514	0.557	0.584	0.647	0.695	0.708	0.651	0.562	0.459	0.424	(40)	
(41)		taud	2.061	1.939	1.809	1.730	1.711	1.614	1.586	1.575	1.652	1.800	1.982	2.051	(41)	
(42)		Ebn at Noon	273	267	257	246	236	219	209	209	220	237	259	268	(42)	
(43)		Edh at Noon	51	60	70	76	76	83	86	88	81	68	55	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1853	2076	2295	2406	2330	2289	2130	2031	2057	1996	1885	1766	(44)	
(45)		RadStd	27	29	48	81	55	54	93	74	63	35	23	23	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page